STAT 405 Project EDA

2026-03-11

```r
suppressPackageStartupMessages(library(dplyr))
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```r
# load in the 2018-2022 dataset
# the file was originally named "British Columbia_Full Data_data.csv"
# (for both the 2018-22 and 2020-24 datasets)
crash18_raw <- read.csv("icbc_crashes_2018_2022.csv", sep="\t", header=TRUE, fileEncoding="utf-16")
crash18 <- crash18_raw |> rename(Derived.Crash.Configuration = Derived.Crash.Congifuration)
```

```r
# change the month from words to numbers
month_names <- c("JANUARY", "FEBRUARY", "MARCH", "APRIL", "MAY", "JUNE",
                 "JULY", "AUGUST", "SEPTEMBER", "OCTOBER", "NOVEMBER", "DECEMBER")

crash18$month <- match(crash18$Month.Of.Year, month_names)
crash18$yearmonth <- crash18$Date.Of.Loss.Year + (crash18$month - 1) / 12

head(crash18)
```

```
##   Crash.Breakdown.2 Date.Of.Loss.Year Animal.Involved Crash.Severity
## 1            SURREY              2018              No CASUALTY CRASH
## 2            SURREY              2018              No CASUALTY CRASH
## 3            SURREY              2018              No CASUALTY CRASH
## 4         COQUITLAM              2018              No CASUALTY CRASH
## 5            SURREY              2018              No CASUALTY CRASH
## 6    PORT COQUITLAM              2018              No CASUALTY CRASH
##   Cyclist.Involved Day.Of.Week Derived.Crash.Configuration Heavy.Veh.Involved
## 1               No      FRIDAY                    REAR END                 No
## 2               No      FRIDAY                    REAR END                 No
## 3               No      FRIDAY                  CONFLICTED                 No
## 4               No      FRIDAY                    REAR END                 No
## 5               No      FRIDAY                    REAR END                 No
## 6               No      FRIDAY                 SIDE IMPACT                 No
##   Intersection.Crash Month.Of.Year Motorcycle.Involved
## 1                 No     SEPTEMBER                  No
## 2                 No          JUNE                  No
## 3                 No         APRIL                  No
## 4                 No       JANUARY                  No
## 5                 No          JUNE                  No
## 6                 No     SEPTEMBER                  No
##   Municipality.Name..ifnull. Parked.Vehicle.Involved Parking.Lot.Crash
## 1                     SURREY                      No                No
## 2                     SURREY                      No                No
```

```
## 3                     SURREY                      No                No
## 4                  COQUITLAM                      No                No
## 5                     SURREY                      No                No
## 6             PORT COQUITLAM                      No                No
##   Pedestrian.Involved         Region Street.Full.Name..ifnull. Time.Category
## 1                  No LOWER MAINLAND           PORT MANN BRIDGE   06:00-08:59
## 2                  No LOWER MAINLAND           PORT MANN BRIDGE   09:00-11:59
## 3                  No LOWER MAINLAND           PORT MANN BRIDGE   12:00-14:59
## 4                  No LOWER MAINLAND                    GLEN DR   12:00-14:59
## 5                  No LOWER MAINLAND           PORT MANN BRIDGE   12:00-14:59
## 6                  No LOWER MAINLAND                BROADWAY ST   12:00-14:59
##   Municipality.Name Road.Location.Description Street.Full.Name Metric.Selector
## 1            SURREY          PORT MANN BRIDGE PORT MANN BRIDGE               1
## 2            SURREY          PORT MANN BRIDGE PORT MANN BRIDGE               1
## 3            SURREY          PORT MANN BRIDGE PORT MANN BRIDGE               1
## 4         COQUITLAM                   GLEN DR          GLEN DR               1
## 5            SURREY          PORT MANN BRIDGE PORT MANN BRIDGE               1
## 6    PORT COQUITLAM               BROADWAY ST      BROADWAY ST               1
##   Total.Crashes Total.Victims Cross.Street.Full.Name Latitude Longitude
## 1             1             1                        49.22589 -122.8180
## 2             1             1                        49.22589 -122.8180
## 3             1             1                        49.22589 -122.8180
## 4             1             1                        49.28274 -122.8063
## 5             1             2                        49.22589 -122.8180
## 6             1             1                        49.24319 -122.7623
##   Mid.Block.Crash Municipality.With.Boundary month yearmonth
## 1               N         COQUITLAM & SURREY     9  2018.667
## 2               N         COQUITLAM & SURREY     6  2018.417
## 3               N         COQUITLAM & SURREY     4  2018.250
## 4               N                  COQUITLAM     1  2018.000
## 5               N         COQUITLAM & SURREY     6  2018.417
## 6               N             PORT COQUITLAM     9  2018.667
```

```r
# load in the 2020-2024 crash dataset
crash20_raw <- read.csv("icbc_crashes_2020_2024.csv", header=TRUE)
crash20 <- crash20_raw
```

```r
crash20$month <- match(crash20$Month.Of.Year, month_names)
crash20$yearmonth <- crash20$Date.Of.Loss.Year + (crash20$month - 1) / 12

head(crash20)
```

```
##   Crash.Breakdown.2 Date.Of.Loss.Year Animal.Flag Crash.Severity Cyclist.Flag
## 1    PORT COQUITLAM              2020          No CASUALTY CRASH           No
## 2   NORTH VANCOUVER              2020          No CASUALTY CRASH           No
## 3         COQUITLAM              2020          No CASUALTY CRASH           No
## 4           LANGLEY              2020          No CASUALTY CRASH           No
## 5         COQUITLAM              2020          No CASUALTY CRASH           No
## 6             DELTA              2020          No CASUALTY CRASH           No
##   Day.Of.Week Derived.Crash.Configuration Heavy.Veh.Flag Intersection.Crash
## 1      FRIDAY SIDE SWIPE - SAME DIRECTION             No                 No
## 2      FRIDAY                UNDETERMINED             No                 No
## 3      MONDAY                    REAR END             No                 No
```

```
## 4      MONDAY                     REAR END             No                 No
## 5      MONDAY                     REAR END             No                 No
## 6      MONDAY                     REAR END             No                 No
##   Month.Of.Year Motorcycle.Flag Municipality.Name..ifnull. Parked.Vehicle.Flag
## 1      DECEMBER              No              PORT COQUITLAM                  No
## 2        AUGUST              No             NORTH VANCOUVER                  No
## 3       JANUARY              No                   COQUITLAM                  No
## 4       JANUARY              No                     LANGLEY                  No
## 5       JANUARY              No                   COQUITLAM                  No
## 6      NOVEMBER              No                       DELTA                  No
##   Parking.Lot.Flag Pedestrian.Flag         Region Road.Location.Description
## 1               No              No LOWER MAINLAND                   UNKNOWN
## 2               No              No LOWER MAINLAND                   UNKNOWN
## 3               No              No LOWER MAINLAND                   UNKNOWN
## 4               No              No LOWER MAINLAND                   UNKNOWN
## 5               No              No LOWER MAINLAND         PORT MANN BRIDGE
## 6               No              No LOWER MAINLAND                   UNKNOWN
##   Street.Full.Name..ifnull. Time.Category Municipality.Name Street.Full.Name
## 1          MARY HILL BYPASS   06:00-08:59    PORT COQUITLAM MARY HILL BYPASS
## 2                     HWY 1   21:00-23:59   NORTH VANCOUVER            HWY 1
## 3                     HWY 1   06:00-08:59         COQUITLAM            HWY 1
## 4                     HWY 1   06:00-08:59           LANGLEY            HWY 1
## 5          PORT MANN BRIDGE   06:00-08:59         COQUITLAM PORT MANN BRIDGE
## 6                TANNERY RD   12:00-14:59             DELTA       TANNERY RD
##   Metric.Selector Total.Crashes Total.Victims Latitude Longitude
## 1               1             1             1       NA        NA
## 2               1             1             3       NA        NA
## 3               1             1             1       NA        NA
## 4               1             1             1       NA        NA
## 5               1             1             1 49.21973 -122.8133
## 6               1             1             1       NA        NA
##   Municipality.With.Boundary Mid.Block.Crash Cross.Street.Full.Name month
## 1                                          N                           12
## 2                                          N                            8
## 3                                          N                            1
## 4                                          N                            1
## 5        COQUITLAM & SURREY                N                            1
## 6                                          N                 HWY 17    11
##   yearmonth
## 1  2020.917
## 2  2020.583
## 3  2020.000
## 4  2020.000
## 5  2020.000
## 6  2020.833
```

```
# filter out 2020-2022 crashes so only 2023-2024 crashes remain
# so we can then combine the two datasets to have 2018-2024 data

crash23 <- crash20 |> filter(yearmonth >= 2023)

# rename columns in crash18 to match those in crash23
crash18 <- crash18|> rename(Animal.Flag = Animal.Involved,
                            Cyclist.Flag = Cyclist.Involved,
```

```
                            Heavy.Veh.Flag = Heavy.Veh.Involved,
                            Motorcycle.Flag = Motorcycle.Involved,
                            Parked.Vehicle.Flag = Parked.Vehicle.Involved,
                            Parking.Lot.Flag = Parking.Lot.Crash,
                            Pedestrian.Flag = Pedestrian.Involved)
```

```r
# combine the crash18 and crash23 data frames
crash <- bind_rows(crash18, crash23)
head(crash)
```

```
##   Crash.Breakdown.2 Date.Of.Loss.Year Animal.Flag Crash.Severity Cyclist.Flag
## 1            SURREY              2018          No CASUALTY CRASH           No
## 2            SURREY              2018          No CASUALTY CRASH           No
## 3            SURREY              2018          No CASUALTY CRASH           No
## 4         COQUITLAM              2018          No CASUALTY CRASH           No
## 5            SURREY              2018          No CASUALTY CRASH           No
## 6    PORT COQUITLAM              2018          No CASUALTY CRASH           No
##   Day.Of.Week Derived.Crash.Configuration Heavy.Veh.Flag Intersection.Crash
## 1      FRIDAY                    REAR END             No                 No
## 2      FRIDAY                    REAR END             No                 No
## 3      FRIDAY                  CONFLICTED             No                 No
## 4      FRIDAY                    REAR END             No                 No
## 5      FRIDAY                    REAR END             No                 No
## 6      FRIDAY                 SIDE IMPACT             No                 No
##   Month.Of.Year Motorcycle.Flag Municipality.Name..ifnull. Parked.Vehicle.Flag
## 1     SEPTEMBER              No                     SURREY                  No
## 2          JUNE              No                     SURREY                  No
## 3         APRIL              No                     SURREY                  No
## 4       JANUARY              No                  COQUITLAM                  No
## 5          JUNE              No                     SURREY                  No
## 6     SEPTEMBER              No             PORT COQUITLAM                  No
##   Parking.Lot.Flag Pedestrian.Flag         Region Street.Full.Name..ifnull.
## 1               No              No LOWER MAINLAND          PORT MANN BRIDGE
## 2               No              No LOWER MAINLAND          PORT MANN BRIDGE
## 3               No              No LOWER MAINLAND          PORT MANN BRIDGE
## 4               No              No LOWER MAINLAND                   GLEN DR
## 5               No              No LOWER MAINLAND          PORT MANN BRIDGE
## 6               No              No LOWER MAINLAND               BROADWAY ST
##   Time.Category Municipality.Name Road.Location.Description Street.Full.Name
## 1   06:00-08:59            SURREY          PORT MANN BRIDGE PORT MANN BRIDGE
## 2   09:00-11:59            SURREY          PORT MANN BRIDGE PORT MANN BRIDGE
## 3   12:00-14:59            SURREY          PORT MANN BRIDGE PORT MANN BRIDGE
## 4   12:00-14:59         COQUITLAM                   GLEN DR          GLEN DR
## 5   12:00-14:59            SURREY          PORT MANN BRIDGE PORT MANN BRIDGE
## 6   12:00-14:59    PORT COQUITLAM               BROADWAY ST      BROADWAY ST
##   Metric.Selector Total.Crashes Total.Victims Cross.Street.Full.Name Latitude
## 1               1             1             1                        49.22589
## 2               1             1             1                        49.22589
## 3               1             1             1                        49.22589
## 4               1             1             1                        49.28274
## 5               1             1             2                        49.22589
## 6               1             1             1                        49.24319
##   Longitude Mid.Block.Crash Municipality.With.Boundary month yearmonth
## 1 -122.8180               N         COQUITLAM & SURREY     9  2018.667
## 2 -122.8180               N         COQUITLAM & SURREY     6  2018.417
## 3 -122.8180               N         COQUITLAM & SURREY     4  2018.250
## 4 -122.8063               N                  COQUITLAM     1  2018.000
## 5 -122.8180               N         COQUITLAM & SURREY     6  2018.417
## 6 -122.7623               N             PORT COQUITLAM     9  2018.667
```

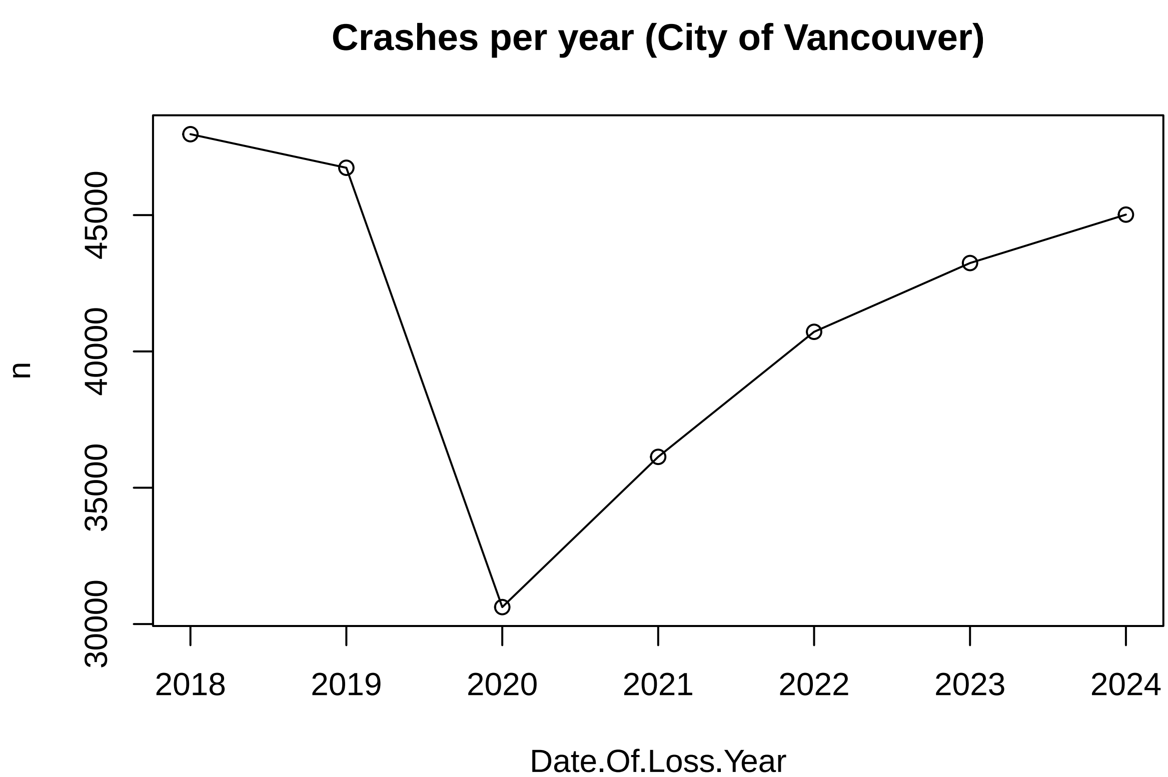
```r
# view crashes only in the City of Vancouver, with known latitude/longitude
# and remove some unnecessary rows

crash_cov <- crash |>
  filter(Municipality.Name == "VANCOUVER") |>
  filter(!is.na(Latitude) & !is.na(Longitude)) |>
  select(-Crash.Breakdown.2, -Region, -Animal.Flag, -Month.Of.Year, -Metric.Selector,
         -Municipality.Name..ifnull., -Municipality.With.Boundary, -month)

head(crash_cov)
```

```
##   Date.Of.Loss.Year Crash.Severity Cyclist.Flag Day.Of.Week
## 1              2018 CASUALTY CRASH           No      FRIDAY
## 2              2018 CASUALTY CRASH           No      FRIDAY
## 3              2018 CASUALTY CRASH           No      FRIDAY
## 4              2018 CASUALTY CRASH           No      MONDAY
## 5              2018 CASUALTY CRASH           No      MONDAY
## 6              2018 CASUALTY CRASH           No      MONDAY
##   Derived.Crash.Configuration Heavy.Veh.Flag Intersection.Crash Motorcycle.Flag
## 1                    REAR END             No                 No              No
## 2                UNDETERMINED             No                 No              No
## 3                    REAR END             No                 No              No
## 4 SIDE SWIPE - SAME DIRECTION             No                 No              No
## 5                    REAR END             No                 No              No
## 6                 SIDE IMPACT             No                 No              No
##   Parked.Vehicle.Flag Parking.Lot.Flag Pedestrian.Flag
## 1                 Yes               No              No
## 2                 Yes               No              No
## 3                 Yes               No              No
## 4                 Yes               No              No
## 5                 Yes               No              No
## 6                 Yes               No              No
##   Street.Full.Name..ifnull. Time.Category Municipality.Name
## 1                W 23RD AVE   15:00-17:59         VANCOUVER
## 2                 CAMBIE ST   15:00-17:59         VANCOUVER
## 3              W HASTINGS ST   21:00-23:59         VANCOUVER
## 4              PRINCESS AVE   15:00-17:59         VANCOUVER
## 5               ORMIDALE ST   18:00-20:59         VANCOUVER
## 6                 W 8TH AVE   18:00-20:59         VANCOUVER
##   Road.Location.Description Street.Full.Name Total.Crashes Total.Victims
## 1                W 23RD AVE       W 23RD AVE             1             1
## 2                 CAMBIE ST        CAMBIE ST             1             1
## 3             W HASTINGS ST    W HASTINGS ST             1             1
## 4              PRINCESS AVE     PRINCESS AVE             1             1
## 5               ORMIDALE ST      ORMIDALE ST             1             1
## 6                 W 8TH AVE        W 8TH AVE             1             1
##   Cross.Street.Full.Name Latitude Longitude Mid.Block.Crash yearmonth
## 1                        49.25004 -123.1114               N  2018.833
## 2                        49.22277 -123.1165               N  2018.833
## 3                        49.28756 -123.1183               N  2018.583
## 4                        49.27821 -123.0916               N  2018.333
## 5                        49.23457 -123.0248               N  2018.417
## 6                        49.26378 -123.1058               N  2018.083
```

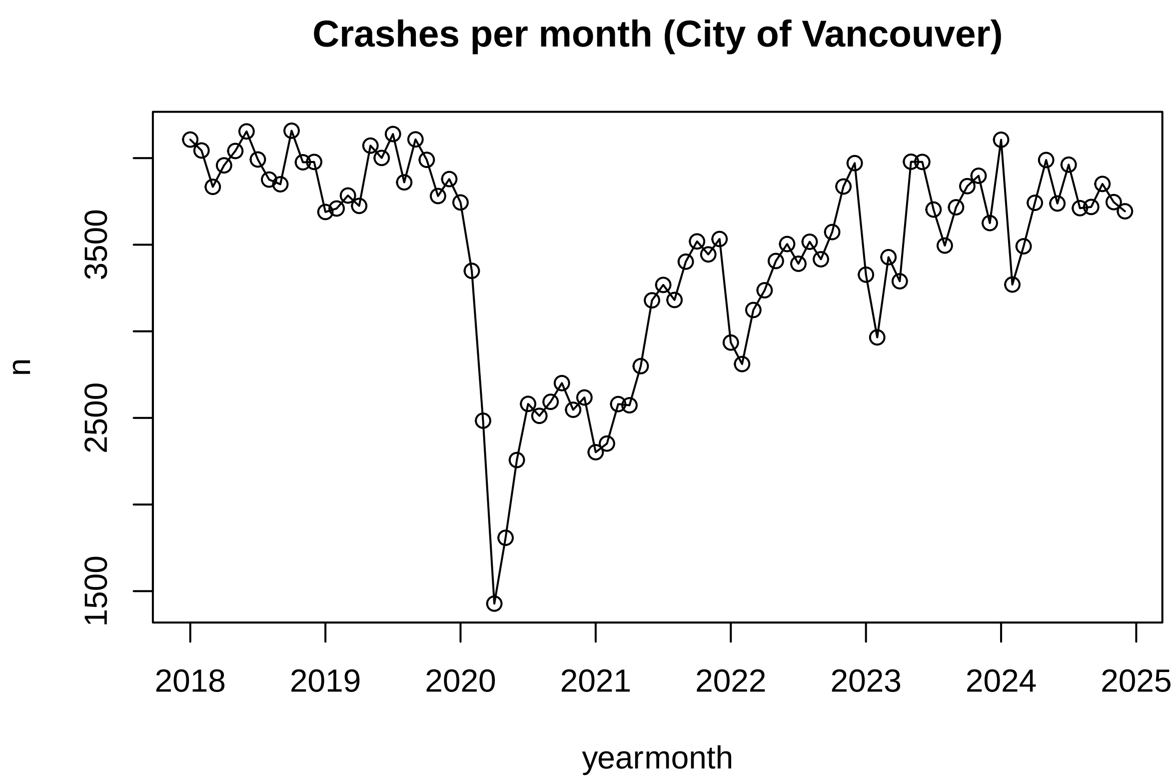
```r
# crashes per year in city of Vancouver
crash_yearly <- crash_cov |> count(Date.Of.Loss.Year)
crash_yearly
```

```
##   Date.Of.Loss.Year     n
## 1              2018 47967
## 2              2019 46736
## 3              2020 30622
## 4              2021 36132
## 5              2022 40719
## 6              2023 43241
## 7              2024 45017
```

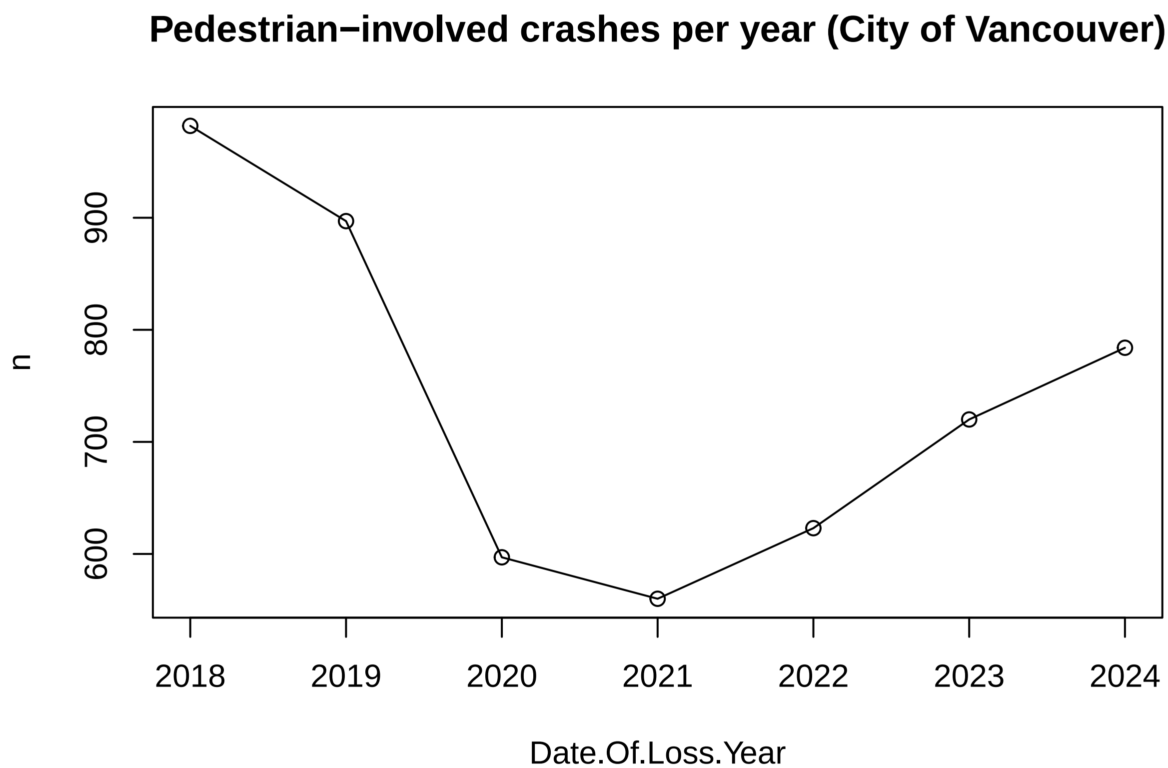
```r
plot(crash_yearly, type="o", main="Crashes per year (City of Vancouver)")
```



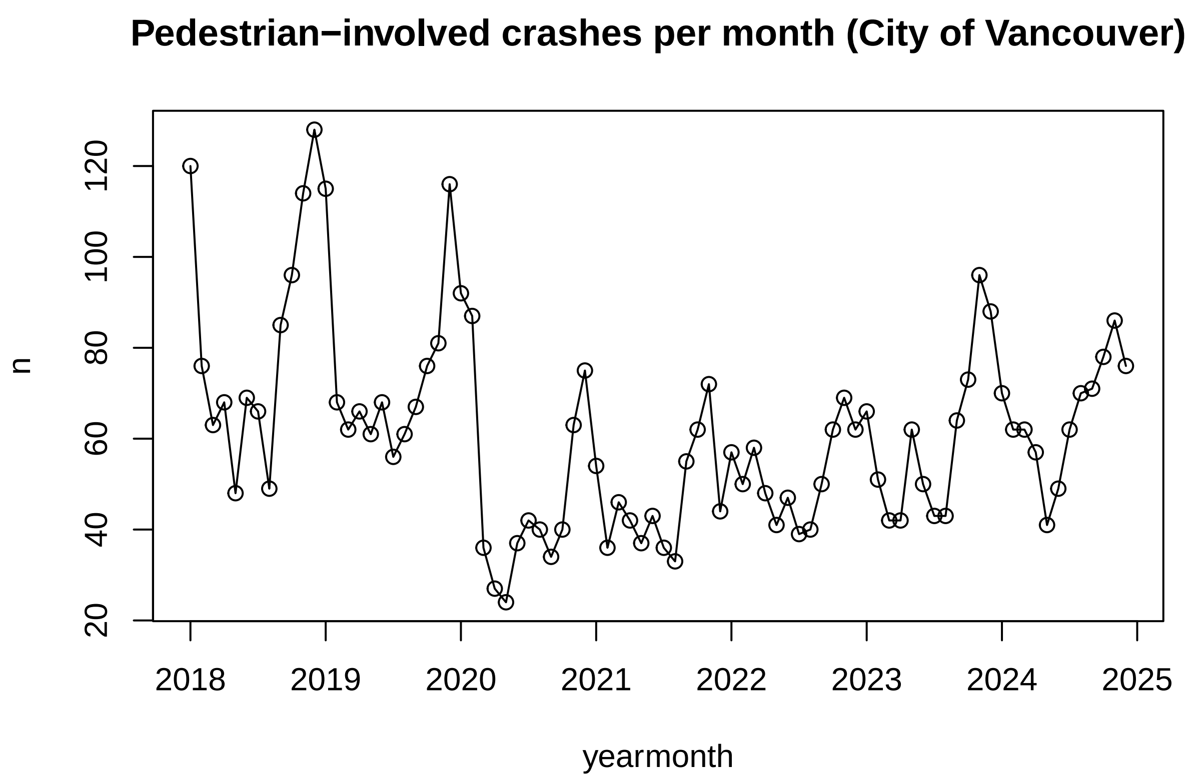
```r
# crashes per month in City of Vancouver, to see if there is seasonality
crash_monthly <- crash_cov |> count(yearmonth)

plot(crash_monthly, type="o", main="Crashes per month (City of Vancouver)")
```



```r
# trends of pedestrian-involved crashes in the city of Vancouver
crash_yearly_ped <- crash_cov |>
  filter(Pedestrian.Flag=="Yes") |>
  count(Date.Of.Loss.Year)

plot(crash_yearly_ped, type="o",
     main="Pedestrian-involved crashes per year (City of Vancouver)")
```



```r
crash_monthly_ped <- crash_cov |>
  filter(Pedestrian.Flag=="Yes") |>
  count(yearmonth)

plot(crash_monthly_ped, type="o",
     main="Pedestrian-involved crashes per month (City of Vancouver)")
```



```r
# weather data
weather18 <- read.csv("yvr_weather_2018.csv", header=TRUE)
weather19 <- read.csv("yvr_weather_2019.csv", header=TRUE)
weather20 <- read.csv("yvr_weather_2020.csv", header=TRUE)
weather21 <- read.csv("yvr_weather_2021.csv", header=TRUE)
weather22 <- read.csv("yvr_weather_2022.csv", header=TRUE)
weather23 <- read.csv("yvr_weather_2023.csv", header=TRUE)
weather24 <- read.csv("yvr_weather_2024.csv", header=TRUE)

weather_dfs <- list(y2018 = weather18,
                    y2019 = weather19,
                    y2020 = weather20,
                    y2021 = weather21,
                    y2022 = weather22,
                    y2023 = weather23,
                    y2024 = weather24)
```

```r
# select columns to keep and create a "yearmonth" column for easier plotting/EDA
# weather[1] is 2018 data, weather[2] is 2019, ... , weather[7] is 2024

for (i in 1:length(weather_dfs)) {
  weather_dfs[[i]] <- weather_dfs[[i]] |>
    select(Date.Time, Year, Month, Day, Max.Temp...C., Min.Temp...C., Mean.Temp...C.,
           Total.Rain..mm., Total.Snow..cm., Total.Precip..mm., Snow.on.Grnd..cm.,
           Dir.of.Max.Gust..10s.deg., Spd.of.Max.Gust..km.h.) |>
    mutate(Spd.of.Max.Gust..km.h. = as.integer(Spd.of.Max.Gust..km.h.),
           yearmonth = Year + (Month - 1) / 12)
}
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `Spd.of.Max.Gust..km.h. = as.integer(Spd.of.Max.Gust..km.h.)`.
## Caused by warning:
## ! NAs introduced by coercion
```

```r
# all weather from 2018 to 2024
weather <- bind_rows(weather_dfs)

head(weather)
```

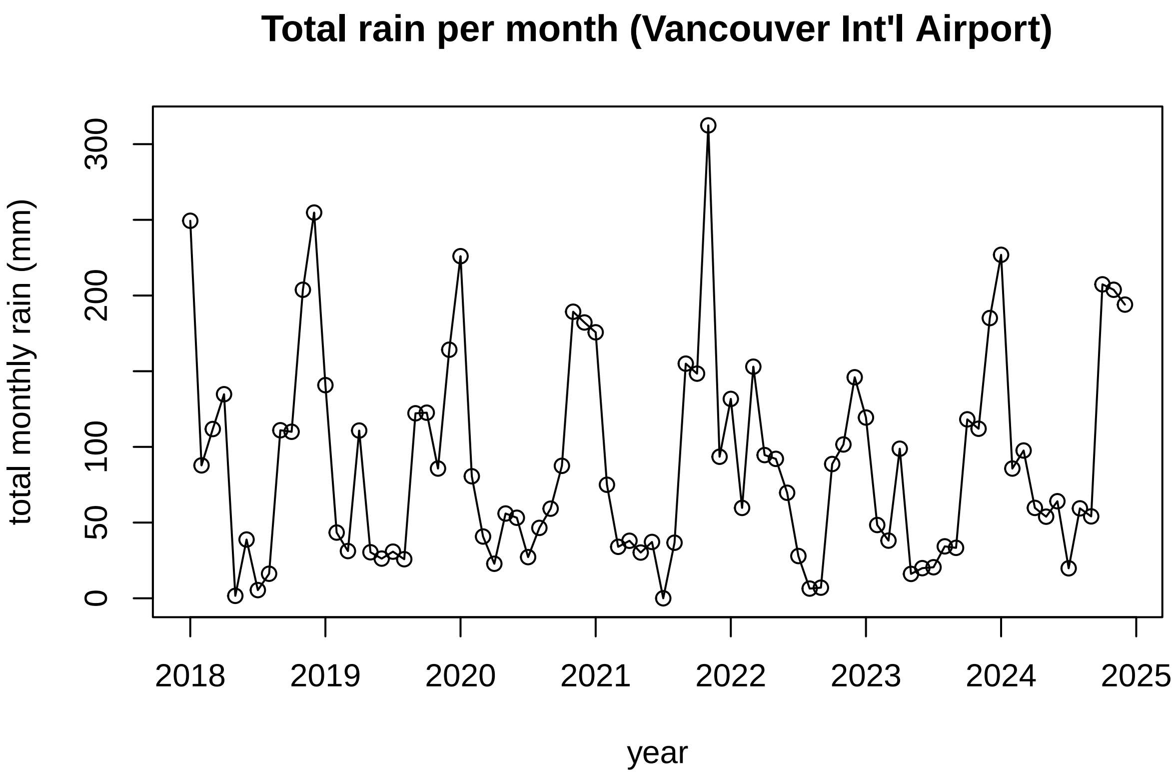
```
##    Date.Time Year Month Day Max.Temp...C. Min.Temp...C. Mean.Temp...C.
## 1 2018-01-01 2018     1   1           3.0          -4.5           -0.8
## 2 2018-01-02 2018     1   2           3.0          -3.3           -0.2
## 3 2018-01-03 2018     1   3           4.0          -1.6            1.2
## 4 2018-01-04 2018     1   4           5.7           1.4            3.6
## 5 2018-01-05 2018     1   5           9.7           4.6            7.2
## 6 2018-01-06 2018     1   6           9.2           5.0            7.1
##   Total.Rain..mm. Total.Snow..cm. Total.Precip..mm. Snow.on.Grnd..cm.
## 1             0.0               0               0.0                NA
## 2             0.0               0               0.0                NA
## 3             0.0               0               0.0                NA
## 4             0.0               0               0.0                NA
## 5            21.8               0              21.8                NA
```

```
## 6             1.0                0               1.0                NA
##   Dir.of.Max.Gust..10s.deg. Spd.of.Max.Gust..km.h. yearmonth
## 1                        NA                     NA      2018
## 2                        NA                     NA      2018
## 3                        NA                     NA      2018
## 4                        NA                     NA      2018
## 5                        NA                     NA      2018
## 6                        29                     43      2018
```

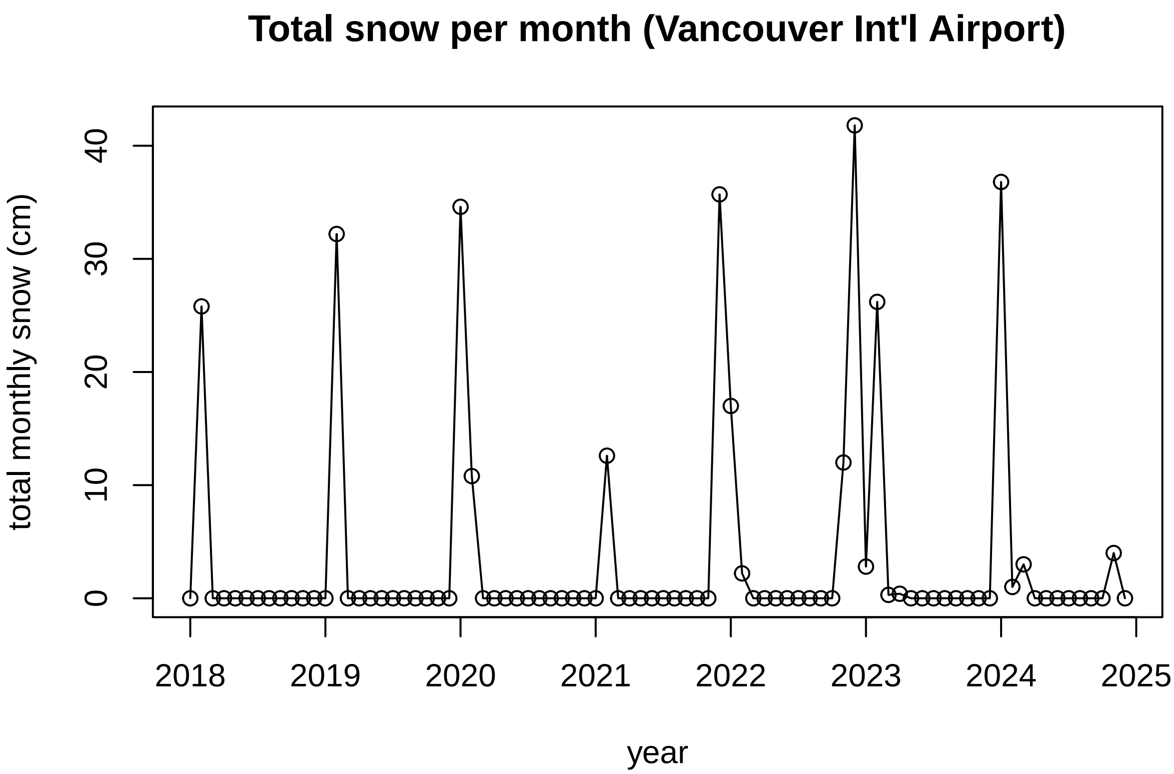
```r
# plot precipitation/temperatures by month

weather_monthly <- weather |>
  group_by(yearmonth) |>
  summarize(monthly_rain_mm = sum(Total.Rain..mm., na.rm=TRUE),
            monthly_snow_cm = sum(Total.Snow..cm., na.rm=TRUE),
            monthly_precip_mm = sum(Total.Precip..mm., na.rm=TRUE),
            avg_max_temp_c = mean(Max.Temp...C., na.rm=TRUE),
            avg_min_temp_c = mean(Min.Temp...C., na.rm=TRUE),
            avg_mean_temp_c = mean(Mean.Temp...C., na.rm=TRUE))

head(weather_monthly)
```

```
## # A tibble: 6 x 7
##   yearmonth monthly_rain_mm monthly_snow_cm monthly_precip_mm avg_max_temp_c
##       <dbl>           <dbl>           <dbl>             <dbl>          <dbl>
## 1     2018            249.              0               249.            7.52
## 2     2018.            87.8            25.8             106.            6.09
## 3     2018.           112.              0               112.            9.54
## 4     2018.           135.              0               135.           12.6
## 5     2018.             1.6             0                 1.6          19.2
## 6     2018.            38.8             0                38.8          19.8
## # i 2 more variables: avg_min_temp_c <dbl>, avg_mean_temp_c <dbl>
```

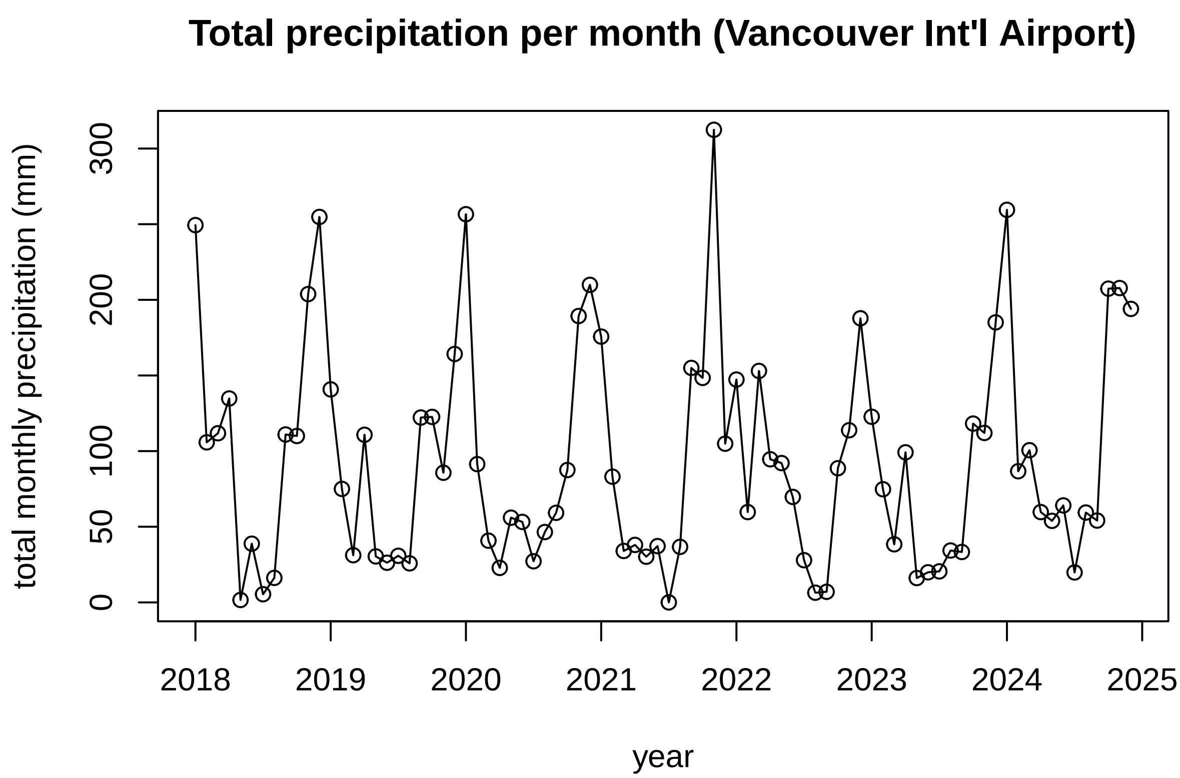
```r
plot(weather_monthly$yearmonth, weather_monthly$monthly_rain_mm, type="o",
     xlab="year", ylab="total monthly rain (mm)",
     main="Total rain per month (Vancouver Int'l Airport)")
```



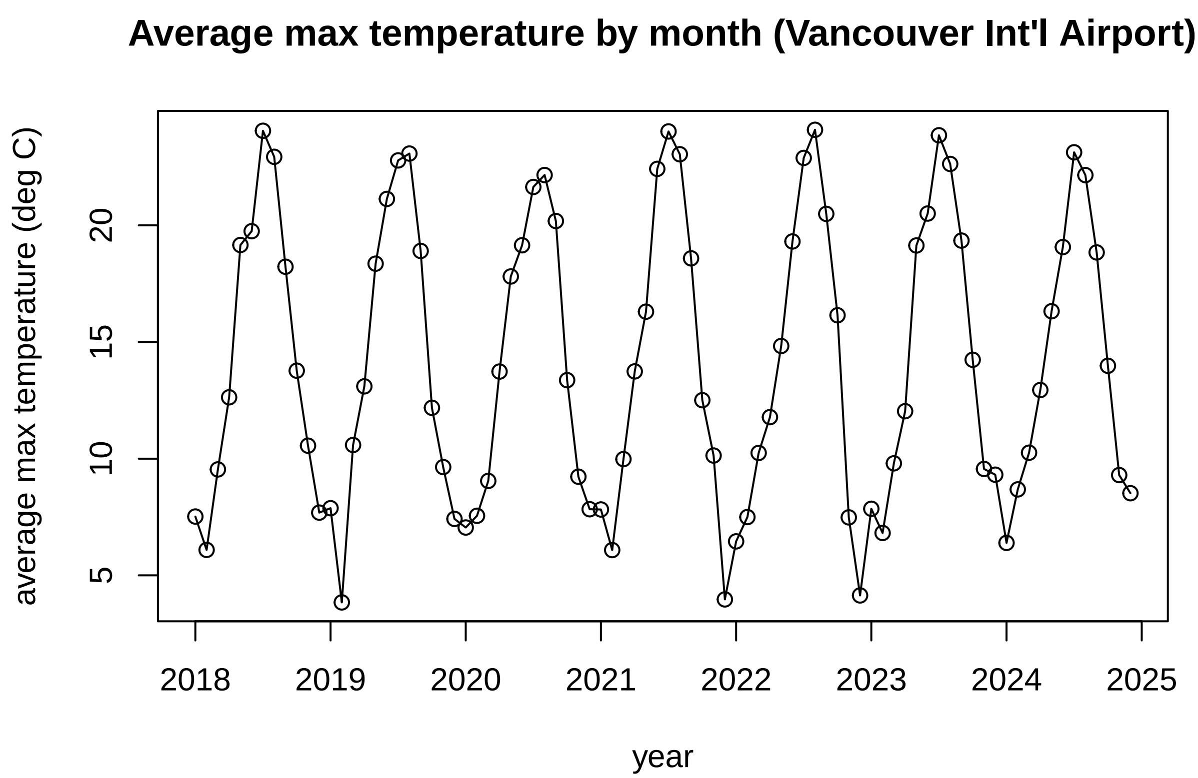
```r
plot(weather_monthly$yearmonth, weather_monthly$monthly_snow_cm, type="o",
     xlab="year", ylab="total monthly snow (cm)",
     main="Total snow per month (Vancouver Int'l Airport)")
```



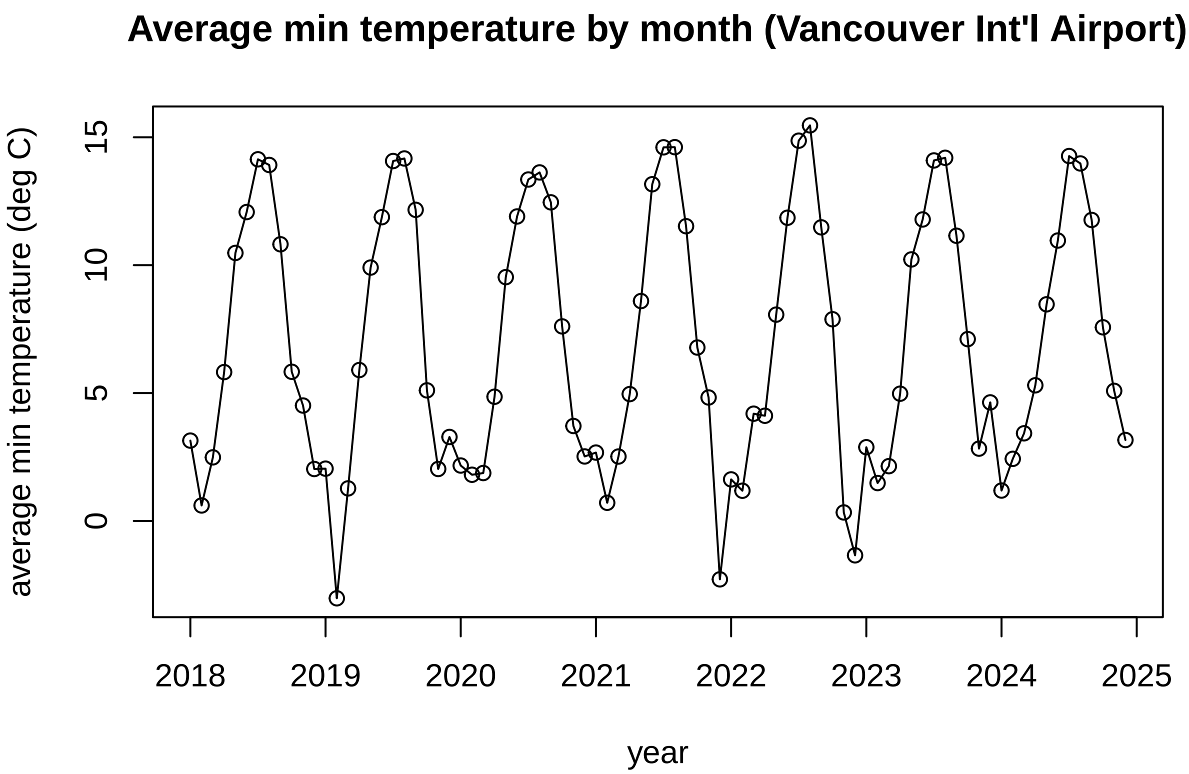
```r
plot(weather_monthly$yearmonth, weather_monthly$monthly_precip_mm, type="o",
     xlab="year", ylab="total monthly precipitation (mm)",
     main="Total precipitation per month (Vancouver Int'l Airport)")
```



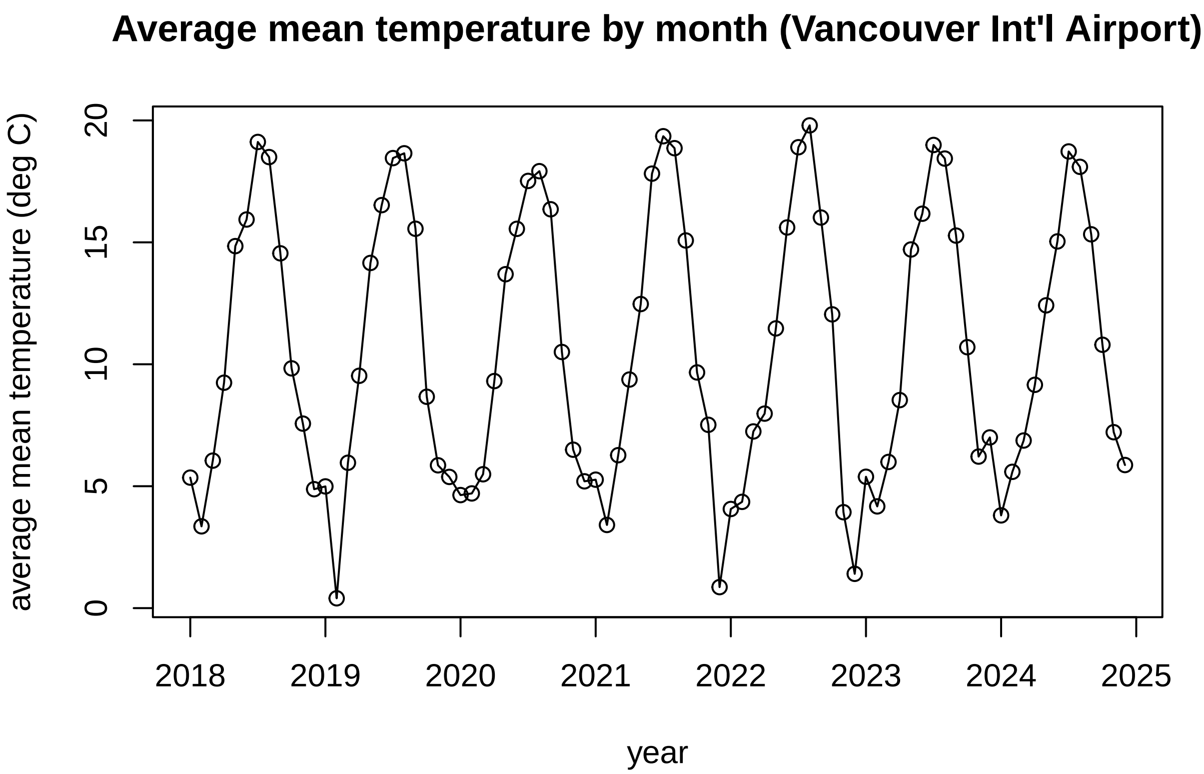
```r
plot(weather_monthly$yearmonth, weather_monthly$avg_max_temp_c, type="o",
     xlab="year", ylab="average max temperature (deg C)",
     main="Average max temperature by month (Vancouver Int'l Airport)")
```



```r
plot(weather_monthly$yearmonth, weather_monthly$avg_min_temp_c, type="o",
     xlab="year", ylab="average min temperature (deg C)",
     main="Average min temperature by month (Vancouver Int'l Airport)")
```



```
plot(weather_monthly$yearmonth, weather_monthly$avg_mean_temp_c, type="o",
     xlab="year", ylab="average mean temperature (deg C)",
     main="Average mean temperature by month (Vancouver Int'l Airport)")
```



```r
# backup dataset

crash_chicago <- read.csv("chicago_crashes.csv", header=TRUE)

crash_chicago <- crash_chicago |>
  select(-CRASH_RECORD_ID, -RD_NO, -CRASH_DATE_EST_I, -LOCATION)

head(crash_chicago)
```

```
##                CRASH_DATE POSTED_SPEED_LIMIT TRAFFIC_CONTROL_DEVICE
## 1 07/12/2023 03:05:00 PM                 30            NO CONTROLS
## 2 07/12/2023 05:50:00 PM                 30            NO CONTROLS
## 3 07/12/2023 02:00:00 PM                 30            NO CONTROLS
## 4 07/12/2023 07:05:00 AM                 30         TRAFFIC SIGNAL
## 5 07/12/2023 06:30:00 PM                 30            NO CONTROLS
## 6 08/05/2022 09:25:00 AM                 30            NO CONTROLS
##   DEVICE_CONDITION     WEATHER_CONDITION LIGHTING_CONDITION FIRST_CRASH_TYPE
## 1      NO CONTROLS                 CLEAR           DAYLIGHT          TURNING
## 2      NO CONTROLS                 CLEAR           DAYLIGHT         REAR END
## 3      NO CONTROLS                 CLEAR           DAYLIGHT     OTHER OBJECT
## 4          UNKNOWN FREEZING RAIN/DRIZZLE           DAYLIGHT         REAR END
## 5      NO CONTROLS                 CLEAR           DAYLIGHT         REAR END
## 6      NO CONTROLS                 CLEAR           DAYLIGHT          TURNING
##   TRAFFICWAY_TYPE LANE_CNT          ALIGNMENT ROADWAY_SURFACE_COND ROAD_DEFECT
## 1     NOT DIVIDED       NA STRAIGHT AND LEVEL              UNKNOWN  NO DEFECTS
## 2     NOT DIVIDED       NA STRAIGHT AND LEVEL                  DRY  NO DEFECTS
## 3     NOT DIVIDED       NA STRAIGHT AND LEVEL                  DRY  RUT, HOLES
## 4     NOT DIVIDED       NA STRAIGHT AND LEVEL              UNKNOWN     UNKNOWN
## 5     NOT DIVIDED       NA STRAIGHT AND LEVEL                  WET  NO DEFECTS
## 6  T-INTERSECTION       NA STRAIGHT AND LEVEL                  DRY  NO DEFECTS
##                REPORT_TYPE             CRASH_TYPE INTERSECTION_RELATED_I
## 1 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 2 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 3 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 4 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 5 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY
## 6 NOT ON SCENE (DESK REPORT) NO INJURY / DRIVE AWAY                      Y
##   NOT_RIGHT_OF_WAY_I HIT_AND_RUN_I        DAMAGE   DATE_POLICE_NOTIFIED
## 1                                    OVER $1,500 07/15/2023 11:30:00 AM
## 2                                  $501 - $1,500 07/12/2023 06:41:00 PM
## 3                  Y               $501 - $1,500 07/21/2023 10:10:00 AM
## 4                                  $501 - $1,500 07/12/2023 02:18:00 PM
## 5                                    OVER $1,500 07/12/2023 07:15:00 PM
## 6                                    OVER $1,500 08/05/2022 10:20:00 AM
##      PRIM_CONTRIBUTORY_CAUSE                SEC_CONTRIBUTORY_CAUSE STREET_NO
## 1 IMPROPER TURNING/NO SIGNAL                   UNABLE TO DETERMINE      4754
## 2      FOLLOWING TOO CLOSELY                        NOT APPLICABLE      8300
## 3             NOT APPLICABLE                        NOT APPLICABLE      9615
## 4      FOLLOWING TOO CLOSELY                   UNABLE TO DETERMINE      2370
## 5      FOLLOWING TOO CLOSELY                 FOLLOWING TOO CLOSELY      5200
## 6 IMPROPER TURNING/NO SIGNAL FAILING TO REDUCE SPEED TO AVOID CRASH      1650
##   STREET_DIRECTION      STREET_NAME BEAT_OF_OCCURRENCE PHOTOS_TAKEN_I
## 1                W         63RD ST                813
```

```
## 2                S       PULASKI RD               834
## 3                S STONY ISLAND AVE               431
## 4                N      ASHLAND AVE              1811
## 5                N       ELSTON AVE              1623
## 6                N        THROOP ST              1433
##   STATEMENTS_TAKEN_I DOORING_I WORK_ZONE_I WORK_ZONE_TYPE WORKERS_PRESENT_I
## 1
## 2
## 3
## 4
## 5
## 6
##   NUM_UNITS      MOST_SEVERE_INJURY INJURIES_TOTAL INJURIES_FATAL
## 1         2 NO INDICATION OF INJURY              0              0
## 2         2 NO INDICATION OF INJURY              0              0
## 3         1 NO INDICATION OF INJURY              0              0
## 4         2 NO INDICATION OF INJURY              0              0
## 5         2 NO INDICATION OF INJURY              0              0
## 6         2 NO INDICATION OF INJURY              0              0
##   INJURIES_INCAPACITATING INJURIES_NON_INCAPACITATING
## 1                       0                           0
## 2                       0                           0
## 3                       0                           0
## 4                       0                           0
## 5                       0                           0
## 6                       0                           0
##   INJURIES_REPORTED_NOT_EVIDENT INJURIES_NO_INDICATION INJURIES_UNKNOWN
## 1                             0                      2                0
## 2                             0                      2                0
## 3                             0                      1                0
## 4                             0                      2                0
## 5                             0                      2                0
## 6                             0                      2                0
##   CRASH_HOUR CRASH_DAY_OF_WEEK CRASH_MONTH LATITUDE LONGITUDE
## 1         15                 4           7 41.77854 -87.74206
## 2         17                 4           7 41.74213 -87.72182
## 3         14                 4           7 41.71984 -87.58479
## 4          7                 4           7 41.92510 -87.66829
## 5         18                 4           7 41.97526 -87.75199
## 6          9                 6           8 41.91222 -87.66191
```